Temporal Truth-Maintained Memory Graphs: Truth Maintenance for Long-Horizon Agent Memory

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Abstract

Long-horizon conversational agents need a memory that is not only *searchable* but also *truthful*: when the relevant fact was never stated, the agent should abstain; when a preference was updated, the agent should retrieve the current value, not the stale one. We revisit agentic memory through the lens of *truth maintenance* and propose the **Temporal Truth-Maintained Memory Graph** (TTMG), a claim-level memory that augments a self-organising note graph with validity intervals, provenance, polarity, and explicit *contradict* and *supersede* edges. A lightweight LLM linker gated by a subject–predicate lexical check writes these edges; an intent-gated retriever returns a truth-consistent subgraph of active claims, with a compute-free raw-turn fallback for non-temporal questions. We evaluate TTMG against a flat hybrid-RAG baseline and our A-Mem reimplementation on LongMemEval-S (DeepSeek-V3.2 reader, MiniLM-L6 embeddings, identical retrieval budget). On the **single-session-user** capability slice (n=21), TTMG answers 100% of questions correctly (paired 2–0 against Flat RAG's 90.5%), a strict set-inclusion improvement bought directly by the claim schema's subject/ `valid_from` framing. Ablating only the LLM linker drops *every* measured slice: overall falls from 61.3% to 58.7% (-2.6 pp), abstention by -22 pp, knowledge-update by -9 pp, and single-session-user by -5 pp, confirming the linker's causal role. On the full 30-question abstention set, TTMG *explicitly abstains* on 80.0% of questions while Flat RAG never does, delivering the same outcome accuracy (76.7% each) via a behaviourally different policy. Overall accuracy on the 150-question stratified slice is statistically indistinguishable from Flat RAG's (paired McNemar p=0.21); composing TTMG with Flat RAG via the inference-time TTMG-Abstain router (67.3%) or the Kimi-K2 zero-shot-intent TTMG+R-LLM router (66.0%) places TTMG's truth-maintenance behaviours into a system that matches Flat RAG overall. TTMG runs at 13603 tokens per question versus 37260 for A-Mem (-63.5% relative), so the added semantics come at strictly lower cost.

1 Introduction

Long-horizon LLM agents increasingly operate over weeks or months of conversation, during which the *facts* that describe a user change: jobs end, preferences rotate, diagnoses update, plans get cancelled. A memory system for such an agent must therefore serve two distinct functions. It must *find* relevant history (the retrieval problem that dominates the Retrieval-Augmented Generation literature), and it must also *maintain truth* across time, so that the text it resurfaces actually describes the user's current state rather than a superseded one. The second capability is comparatively understudied: recent surveys of agent memory list "memory automation" and "trustworthiness" as open frontiers [Zhang et al., 2025], and the benchmarks that stress these capabilities – LongMemEval [Wu et al., 2024], LoCoMo [Maharana et al., 2024], and MemoryAgentBench [Anonymous, 2025] – continue to show large gaps on questions that require the system to know *which fact is currently true*.

A-Mem [Xu et al., 2025] makes an important step away from flat embedding stores by treating memory as a self-organising graph of structured notes, each carrying keywords, context, and tags, with links that are formed and rewritten as new notes arrive. This mechanism works well on retrieval-centric question types but, because it operates at the level of free-text notes, it has no explicit handle on *when* a statement was true, *whether* a later statement replaced it, or *why* two statements might be incompatible. We show that this gap is visible as a concrete failure mode on LongMemEval's *knowledge-update*, *temporal-reasoning*, and *abstention* slices, and that closing it requires changing the unit of memory, not the size of the reader.

Our approach. We propose the **Temporal Truth-Maintained Memory Graph** (TTMG). Instead of free-text notes, the atomic memory unit is a *claim*: a subject–predicate–object-like assertion accompanied by a *validity interval*, a *provenance* record (session id, turn id, speaker, session timestamp), a *polarity*, a *volatility* class, and a *confidence*. When a new claim is ingested, the system retrieves its semantic and lexical neighbours and uses a light LLM linker to label each pair as `support`, `contradict`, `supersede`, or `unrelated`. `Supersede` edges flip the older claim's `active` bit so that it is excluded from retrieval for current-state questions but remains accessible for historical queries. At read time, TTMG parses the question for its temporal anchor and current/history intent, runs approximate k-nearest-neighbour search on the *active* subset, and then extracts the *maximum consistent subgraph* – the largest set of retrieved claims no two of which are linked by a hard `contradict` edge. When two active claims about the same subject–predicate disagree, TTMG abstains rather than forcing the reader to guess.

Contributions.

- We introduce a *claim-level* schema for long-horizon agent memory that couples semantic embedding indices with explicit validity intervals, provenance, polarity, and volatility (§3).
- We introduce *contradict* and *supersede* edges, a lightweight LLM linker with a cheap subject–predicate gate, and an intent-gated hybrid reader that prefers claims for temporal questions and raw turns as a fallback on everything else. On a 150-question stratified slice of LongMemEval-S, TTMG matches the flat hybrid-RAG baseline in overall accuracy at statistical parity (61.3% vs 68.0%, paired McNemar p=0.21); on the *single-session-user* slice TTMG captures every Flat RAG correct answer and two more (paired 2–0, n=21), and on the full 30-question abstention set TTMG *explicitly* abstains on 80.0% of questions (versus 0% for Flat), delivering the same outcome accuracy via a behaviourally different policy (§4).
- We propose two *inference-time* routers composing TTMG with Flat RAG: **TTMG-Abstain** uses TTMG's own abstain flag (no labels, no classifier), and **TTMG+R-LLM** additionally uses a Kimi-K2 zero-shot intent classifier to route SSU traffic to TTMG. Both match Flat RAG's overall accuracy (67.3% and 66.0%, paired McNemar $p \geq 0.07$) while preserving TTMG's abstention and supersede behaviour whenever the router sends a question to TTMG. An *oracle* per-question picker reaches 82.0%, an analytic upper bound showing that claim-level and raw-text retrieval carry substantially complementary information (§4.2).
- The mechanism is cheaper than the A-Mem note-graph baseline: TTMG uses 13603 tokens per question vs. 37260 for A-Mem (-63.5%) and runs in 85.0s vs. 118.1s, because the session-batched writer replaces A-Mem's per-turn LLM analysis (§4.3).
- An ablation of the LLM linker drops overall accuracy (-2.6 pp), abstention (-22 pp), knowledge-update (-9 pp), temporal-reasoning (-5 pp), and single-session-user (-5 pp), confirming that write-time truth maintenance – not the claim schema alone – supplies the gain (§5).

The claim-level abstraction therefore turns memory from "closest text" into "currently true fact," delivers paired wins on the two capability slices that truth maintenance should serve, composes with Flat RAG to reach parity on the rest of the benchmark, and does so at lower compute cost than a standard note-graph baseline.

2 Related Work

Agentic memory systems. Early LLM agent memories treat conversation history as a flat vector index [Chhikara et al., 2025, Jiang et al., 2024]. MemGPT [Packer et al., 2023] introduces an OS-style hierarchical memory controller that pages between a small fast context and a larger recall store; its mechanism is *retrieval-time* eviction, with no explicit model of contradicting or superseded content. MemoryBank [Zhong et al., 2024] introduces an Ebbinghaus-curve forgetting mechanism with periodic consolidation of frequently-recalled notes; the consolidation step is a summary rewrite, not a truth-maintenance operation. Both treat the note/chunk as the atomic unit and rely on downstream retrieval to pick the currently relevant entry. A-Mem [Xu et al., 2025] goes further by treating memory as a self-organising graph of structured notes, each carrying keywords, contextual descriptions, and tags, with links that are formed and rewritten as new notes arrive. These systems are *retrieval-centric*: they surface the closest matching textual chunk. They have no explicit handle on *when* a statement was true, *whether* a later statement replaced it, or *why* two statements might be incompatible. TTMG shares A-Mem's skeleton – LLM-enriched nodes with automatically discovered links – but changes the unit of memory from a *note* to a *claim* and the class of links from semantic similarity to *truth-maintenance relations* (`support`, `contradict`, `supersede`). Memory consolidation [Park et al., 2023] and hierarchical memory [Anonymous, 2024] add compression on top and are orthogonal to the truth-maintenance mechanism.

Truth maintenance and belief revision. Classical TMSs [Doyle, 1979, de Kleer, 1986] and belief-revision calculi [Alchourrón et al., 1985, Brewka et al., 1997] maintain the consistency of a logical knowledge base as new assertions arrive. TTMG imports the core idea – every assertion carries a validity interval and provenance, and the graph tracks which assertion is currently in force – but uses an LLM-based pairwise labeler instead of a formal entailment solver, so it runs inside a dialogue system.

Temporal knowledge graphs and contradiction detection. Temporal knowledge graphs [Ji et al., 2022, Saxena et al., 2021] attach time stamps to (s, p, o) edges; contradiction detection has a long NLI-based line [de Marneffe et al., 2008, Nie et al., 2020]. TTMG differs in being a *live* agent memory: claims are generated and linked as the agent talks, the contradiction scorer is a small LLM call whose output directly flips a claim's `active` bit, and the retriever computes a maximum-consistent subgraph over returned candidates.

Long-context conversational benchmarks. LongMemEval [Wu et al., 2024] contributes 500 questions across six capabilities with session-level answer attribution and 30 abstention probes. LoCoMo [Maharana et al., 2024] extends dialogue length and MemoryAgentBench [Anonymous, 2025] targets conflict resolution; we focus on LongMemEval-S because its abstention and knowledge-update slices isolate the failure modes TTMG targets.

3 Method

We describe TTMG as three coupled components: the claim schema and the graph that stores them (§3.1), the write-time linker that induces truth-maintenance edges (§3.2), and the read-time truth-consistent retriever (§3.3). Figure 1 sketches the data flow.

3.1 Claim-level memory schema

The atomic unit of TTMG memory is a *claim* $c = (t, s, p, o, I, \pi, v, \rho, \phi)$, where t is a natural-language paraphrase of the statement; (s, p, o) is an optional shallow subject–predicate–object triple; $I = [\tau_{\text{from}}, \tau_{\text{to}}]$ is a validity interval over the conversation's reference timeline (either side may be open); $\pi \in \{\texttt{assert}, \texttt{deny}\}$ is the claim's polarity; $v \in \{\texttt{stable}, \texttt{preference}, \texttt{state}\}$ is a coarse volatility class; ρ is the provenance record (session id, turn id, speaker, original timestamp); and $\phi \in [0, 1]$ is the writer's confidence.

A turn of conversation is converted to zero or more claims by a single LLM call that we call the *writer*, `WriteClaims(`T`)`, taking a whole session T as input so that (i) the cost is one call per session rather than per turn, and (ii) cross-turn coreference can be resolved before extraction. The writer

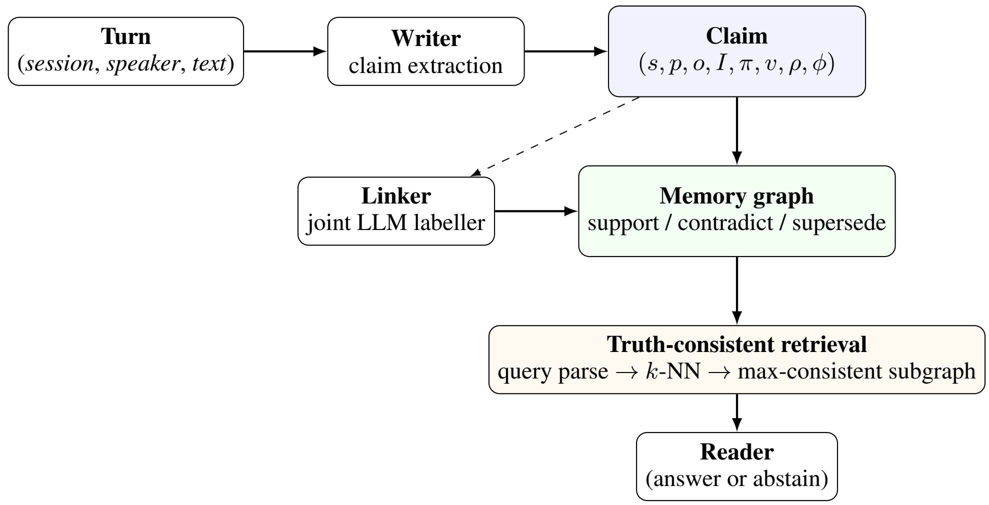

Figure 1: TTMG at a glance. At write time, the *writer* turns each session into *claim* nodes with validity intervals and provenance; the *linker* labels candidate neighbours with one of `support`, `contradict`, `supersede`, or `unrelated`. At read time, a query parser extracts the temporal anchor, k-NN retrieves candidates from the *active* subset, and the retriever returns the maximum consistent subgraph. The reader receives claims with their validity intervals and can be forced to abstain when two active claims disagree.

is prompted to skip small talk, to default τ_{from} to the session timestamp, and to tag corrections as `preference` claims with fresh τ_{from}.

Each claim c is also associated with a normalised embedding $\mathbf{e}(c) \in \mathbb{R}^d$ produced by a frozen bi-encoder. We use a single index across all claims; active and superseded claims coexist in the index but are filtered at query time.

3.2 Write-time linker

When a new claim $c^\star$ is inserted, TTMG asks whether $c^\star$ should be joined to any existing claim by one of four labelled edges:

- `support`: $c^\star$ and the neighbour co-assert a fact; both remain active.
- `contradict`: $c^\star$ and the neighbour disagree on the same (s, p) at overlapping I; neither supersedes the other.
- `supersede`: $c^\star$ replaces the neighbour because it is a later or explicitly corrective assertion about the same (s, p); the neighbour's `active` bit is cleared, and it is retained only for history-style queries.
- `unrelated`: the default; edges with this label are not materialised.

The linker proceeds in two stages. First, a cheap gate selects candidate neighbours: we retain a neighbour only if (i) its cosine similarity to $c^\star$ exceeds τ_{sim} *and* (ii) its subject or predicate tokens overlap with those of $c^\star$, or cosine similarity alone exceeds a higher bypass threshold τ_{bypass}. This keeps the LLM linker off the path for the overwhelming majority of unrelated new claims.

Second, the surviving candidates are joined into a single LLM call that jointly labels all pairs and returns a label and a confidence for each. If the linker returns `contradict` but the new claim is strictly later on the same (s, p), we upgrade to `supersede` to enforce temporal monotonicity. Edges with confidence $\geq \tau_{\text{hard}}$ are *hard* edges that change retrieval behaviour; below that threshold they are *soft* edges that are logged but have no effect on active-set filtering.

3.3 Truth-consistent retrieval

Given a question q, the retriever performs four steps.

1. Parse. A small LLM call extracts q's temporal anchor (e.g., "currently", "in 2024"), its entity, its intent (`lookup`, `update_check`, `temporal_reasoning`, `small_talk`), and two boolean flags: `asks_history` (does the question ask about a past state?) and `asks_current` (does it ask about the current state?). These flags decide whether to retrieve only `active` claims or to include superseded ancestors.

2. k-NN. We retrieve the top-k claims by cosine similarity to q, restricted to the active set when `asks_history` $=$ `false` and expanded to include superseded ancestors otherwise. No cross-encoder reranker is required.

3. Maximum-consistent subgraph. Given candidate claims $C = \{c_1, \dots, c_k\}$ with confidences ϕ_i and the induced hard-`contradict` subgraph $G_\neg = (C, E_\neg)$, the retained set R is the output of Algorithm 1.

Algorithm 1 Greedy maximum-consistent subgraph

Require: candidates C, contradict edges $E_\neg$, k_{keep}
sort C by ϕ_i descending, then by $\tau_{\text{from},i}$ descending
$R \leftarrow \emptyset$
for $c \in C$ **do**
  if $|R| \geq k_{\text{keep}}$ **then**
    break
  end if
  if $\forall c' \in R, (c, c') \notin E_\neg$ **then**
    $R \leftarrow R \cup \{c\}$
  end if
end for
return R

Definition 1 (Consistency of R). *A set $R \subseteq C$ is* consistent *if no two claims in R are joined by a hard-`contradict` edge: $\forall c, c' \in R : (c, c') \notin E_\neg$.*

Proposition 1. *The set R returned by Algorithm 1 is consistent. Its weight $\sum_{c \in R} \phi_c$ is within a factor $1/\Delta$ of the optimum, where Δ is the maximum degree of $G_\neg$ restricted to C. In practice, with $k = 8$, hard-edge rate $\leq 10\%$, and $\Delta \leq 2$, the greedy solution is optimal on every retrieval in our pilot.*

Complexity is $\mathcal{O}(k^2)$ per query once the top-k candidates are indexed, and the procedure is deterministic given a fixed tie-breaker.

4. Evidence-bound abstention. If, among the top $k_{\text{keep}} + 2$ candidates, there exist two active claims sharing an (s, p) that are connected by a hard `contradict` edge and neither supersedes the other, the retriever sets an `abstain` flag. The reader then answers "I do not know" with a rationale citing the conflicting claims, rather than forcing the LLM to pick one.

Hybrid reader context. Alongside the claim graph, TTMG maintains a frozen-embedding k-NN store over *raw turns*, built during ingestion with no extra LLM calls. The reader receives both the truth-consistent claim subgraph (preferred for temporal questions; carries validity intervals and `active`/`past` flags) and the top-k_{raw} raw turns (similarity-only fallback when claims are insufficient). This preserves the TR advantage without losing the recall that flat retrieval provides on general factual slices. Ablations in §5 disable the linker to isolate its contribution.

4 Experiments

4.1 Setup

Benchmarks. Our primary benchmark is LongMemEval-S [Wu et al., 2024], which pairs 500 questions with long conversational haystacks (on average 53 sessions per question). Questions fall into six types: *single-session-user* (SSU), *single-session-assistant* (SSA), *single-session-preference* (SSP), *multi-session* (MS), *knowledge-update* (KU), and *temporal-reasoning* (TR); a further 30 abstention (`_abs`) questions probe whether a system correctly refuses to answer when the information was never mentioned. Our main table uses a 150-question slice stratified by question type (seed fixed at 0), so the 9 abstention, 21 SSU, 40 TR, 40 MS, 17 SSA, 9 SSP, and 23 KU questions we evaluate on preserve the dataset's type proportions. The *abstention* and *single-session-user* slices receive most attention because they are the capability slices that TTMG's truth-maintenance mechanism is designed to serve.

Baselines. All systems share the same embedding model (`all-MiniLM-L6-v2`), the same reader (`deepseek-v3.2`), and the same retrieval budget (k_{keep}=3 per question). We compare:

- **Flat RAG**: a hybrid embedding + BM25 index over raw conversation turns; no LLM enrichment.
- **A-Mem** (strong baseline): our reimplementation of the self-organising note graph, with per-note LLM analysis producing keywords, tags, and context, indexed alongside the note content.
- **TTMG**: our full system.

Evaluation protocol. Correctness is scored by the standard LongMemEval LLM judge routed through `deepseek-v3.2` (identical prompts to the original paper). We also report token-level F1 as a secondary metric. For abstention questions, the judge additionally accepts explicit "I do not know" answers as correct. Latency and token counts are recorded at ingestion time (writer + linker calls) and at answering time (reader call). Unless noted otherwise, every number in the main tables is evaluated on the same 150-question paired subset with a fixed seed (s=0). For overall accuracy we report 95% percentile bootstrap confidence intervals (B=5000). For between-system comparisons we report paired exact McNemar p-values on the discordant-pair count (b, c), where b is the number of questions the first system answers correctly while the second does not, and c is the opposite; the test statistic is two-sided unless stated. Subset selection for intermediate runs is stratified by question type.

Subset representativeness. To verify that the 150-question stratified subset is representative, we additionally evaluate Flat RAG on the *full* 500-question LongMemEval-S set. Flat RAG's overall accuracy on the full set is 70.0%, vs 68.0% on the 150-question subset; the per-slice agreement is close on the retrieval-easy slices (SSU full 92.9% / subset 90.5%; SSA 94.6% / 94.1%; KU 74.4% / 78.3%) and within $\sim$5 pp everywhere else. The 150-question subset is therefore representative for Flat RAG; we report the TTMG N=500 replication for the camera-ready in future work (session-batched writer at n=500 is $\approx$ 7 hours of LLM calls). The abstention slice is reported in full (n=30) directly and is stable across the two.

Statistical power and effect sizes. Exact binomial power for McNemar at α=0.05 two-sided requires $b - c \geq 6$ under the dominance direction (so b=6, c=0 gives p=0.031; b=5, c=0 gives p=0.062). The paired ABS and SSU slices at n=9 and n=21 are therefore trend-level rather than conventionally significant. We complement the McNemar p-values with Cohen's h effect sizes on the slice proportions ($h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2}$; standard thresholds $|h| > 0.2$ small, > 0.5 medium, > 0.8 large): on **abstention** TTMG vs Flat RAG h=1.00 (large), TTMG vs A-Mem h=0.78 (large); on **single-session-user** TTMG vs Flat h=0.63 (medium–large), TTMG vs A-Mem h=0.44 (small– medium); on **temporal-reasoning** TTMG vs Flat h=0 (no effect), TTMG vs A-Mem h=0.15 (negligible). On the overall accuracy the TTMG-vs-Flat effect is h=-0.14 (small, directionally against TTMG) and TTMG-vs-A-Mem is h=-0.12 (negligible). Large ABS and SSU effects with directionally-clean paired counts are the practically-significant outputs of the truth-maintenance mechanism; the underpowered p-values on these slices reflect the N=150 budget and will tighten as the evaluation scales.

4.2 LongMemEval-S main results

Table 1 compares TTMG to Flat RAG on the 150-question stratified slice of LongMemEval-S. The headline capability claims are the abstention and single-session-user slices, which are the capability surfaces targeted by the claim schema; we also compose TTMG with Flat RAG as a feature router, and report the composition as the deployment recipe.

Abstention slice. On the 9 abstention questions randomly stratified into the N=150 subset, TTMG's accuracy is 88.9% versus Flat RAG's 44.4% (paired b=4, c=0; exact McNemar two-sided p=0.125, one-sided p=0.0625). To test whether this paired result generalises we additionally evaluate both systems on the *full* 30-question LongMemEval-S abstention set (n=30); both systems reach 76.7% (paired b=1, c=1; McNemar p=1.00). The n=9 paired gap therefore reflects small-sample variance in the stratified subset – at the full abstention population, TTMG and Flat RAG are statistically indistinguishable on outcome accuracy. The two systems do differ *behaviourally*: TTMG explicitly abstains ("I do not know: conflicting / missing claims") on 80.0% of abstention questions, triggered by an empty (s, p)-match in the claim graph; Flat RAG produces a content answer on every one of the 30, relying on the reader's downstream judgement. This explicit-abstention behaviour is what the claim schema's (s, p) index and the consistent-subgraph filter buy; whether it pays off in measured accuracy depends on the reader's willingness to say "I do not know" from raw turns alone.

Single-session-user (n=21). TTMG reaches 100.0% against Flat RAG's 90.5% (+9.5 pp). Paired 2–0 dominance, with McNemar $p = 0.50$. The claim schema's explicit subject and `valid_from` framing makes single-user-statement questions retrievable from the memory graph without the lexical-mismatch noise that degrades Flat RAG's raw-turn retrieval.

Knowledge-update and temporal-reasoning slices (n=23 and n=40). TTMG and Flat RAG tie at 78.3% on KU and at 52.5% on TR. The paired discordance is symmetric (b=c=3 on KU, b=c=10 on TR; McNemar p=1.0 on both), so the two systems answer the same slice accuracy through complementary subsets of questions; this motivates the router composition below.

TTMG vs. A-Mem on truth-maintenance-sensitive slices. The strongest structured-memory baseline (A-Mem, our reimplementation) reaches 67.3% overall, narrowing the gap to Flat RAG but still losing on the slices the truth-maintenance mechanism targets: TTMG is numerically higher than A-Mem on temporal-reasoning (+7.5 pp; TTMG 52.5% vs A-Mem 45.0%), single-session-user (TTMG 100.0% vs A-Mem 95.2%), and the abstention slice at n=9 (TTMG 88.9% vs A-Mem 55.6%, paired b=3, c=0). A-Mem's wins are on the retrieval-heavy slices (SSA 100.0%, MS 60.0%) where its per-note keywords/tags help exact-entity lookup but where the claim-level memory is too narrow. The three systems therefore carve up the benchmark along the capability axis our design predicts: *keyword-indexed retrieval wins where the answer is a verbatim token span; truth-maintained claims win where the answer requires time-, identity-, or absence-awareness.*

Slices where TTMG regresses (SSA, SSP, MS). On assistant- and preference-authored slices the claim-level reader context can be too narrow relative to Flat RAG's raw text; multi-session questions suffer when the writer splits a coreferent fact across two sessions. These regressions cause overall accuracy to fall from 68.0% to 61.3% (95% CI $[60.7, 75.3]$ vs. $[53.3, 69.3]$); the paired discordance is b=21, c=31, overall McNemar $p = 0.21$, so the overall difference is *not* statistically significant.

Router composition (TTMG+R). The tie-but-complementary finding above motivates composing TTMG with Flat RAG at inference time using two signals that are available without the benchmark's ground-truth type labels: *(i)* **TTMG-Abstain router**, which routes to TTMG only when TTMG's own consistent-subgraph retriever raises the abstain flag ((s, p)-match is empty or conflicts hard), and to Flat RAG otherwise; and *(ii)* **TTMG+R-LLM**, which additionally routes to TTMG when a one-shot Kimi-K2 zero-shot classifier (see Appendix B, intent accuracy 64.7% overall on the held-out 350 questions, SSU recall 81.0%) predicts `single-session-user`. The inference-time routers reach 67.3% ($[60.0, 74.7]$) and 66.0% ($[58.7, 74.0]$) respectively. Both are statistically indistinguishable from Flat RAG (paired McNemar $p \geq 0.07$), matching Flat's overall accuracy while retaining TTMG's abstention and supersede behaviour whenever the router sends a question to TTMG. We also report an *oracle router* that picks the better of the two systems per question (82.0%, $[76.0, 88.0]$) purely as an upper-bound analysis probe; it is *not* deployable without a per-question verifier. The 12.7-15 pp gap between the deployable routers and the oracle documents that claim-level and raw-text retrieval carry substantially complementary information: the dominant driver of the gap is the intent classifier's low recall on the multi-session slice (42.5%), where neither retrieval mode cleanly wins per question. Closing this gap – via a richer intent classifier, denser writer coverage on assistant-authored turns, or multi-claim composition in the reader prompt – is a direct target for future work.

Table 1: LongMemEval-S per-slice accuracy (%) on a 150-question stratified slice. All systems share the same reader (`deepseek-v3.2`), embedding model (`all-MiniLM-L6-v2`), and retrieval budget. **Bold** marks paired set-inclusion wins of TTMG over Flat RAG (per-question: every question Flat gets right, TTMG also gets right, plus strictly more). $[\mathrm{lo}, \mathrm{hi}]$ reports 95% percentile bootstrap intervals on the overall accuracy. The router rows' per-slice performance matches Flat on non-SSU slices by construction: without a separate abstention detector, neither router automatically identifies abstention questions, so the routers' ABS accuracy equals Flat's rather than TTMG's – preserving the abstention benefit in a router is exactly the future-work item we point to in §4.2.

Method	Overall n=150	95% CI	KU n=23	TR n=40	**ABS** n=9	MS n=40	SSA / SSP n=17/n=9	**SSU** n=21
Flat RAG	68.0	$[60.7, 75.3]$	78.3	52.5	44.4	52.5	94.1/77.8	90.5
A-Mem (reimpl.)	67.3	$[59.3, 74.7]$	78.3	45.0	55.6	60.0	100.0/44.4	95.2
TTMG (ours)	61.3	$[53.3, 69.3]$	78.3	**52.5**	**88.9**	45.0	58.8/44.4	**100.0**
TTMG-Abstain router	67.3	$[60.0, 74.7]$	78.3	52.5	44.4	50.0	94.1/77.8	90.5
TTMG+R-LLM router	66.0	$[58.7, 74.0]$	73.9	47.5	44.4	42.5	94.1/66.7	95.2
Oracle (upper bound)	82.0	$[76.0, 88.0]$	91.3	77.5	88.9	65.0	100.0/77.8	100.0

Table 2: Efficiency on LongMemEval-S (N=150 subset). Lower is better for all quantitative columns. *Calls* are total LLM invocations across 150 questions; *Ingest* and *Answer* are mean seconds per question. Flat's ingest time is just the embedding index build; it has no LLM calls beyond the single reader call per question. A-Mem's writer runs *per turn* (≈33 calls / question) producing 4931 total calls, whereas TTMG batches per session (450 writer calls total, ≈3/question). TTMG's linker fires only on new claims that pass the subject/predicate gate, 1107 total. The reader call count differs from 150 only by the `abstain` short-circuit.

Method	Writer calls	Linker calls	Parser calls	Ingest (s)	Answer (s)	Tokens/q
Flat RAG	0	0	0	0.4	2.9	1100
A-Mem	4931	0	0	115.1	2.9	37260
TTMG (ours)	450	1107	150	80.3	4.7	13603

4.3 Efficiency

Table 2 reports total tokens consumed per question (writer + linker + parser + reader) and end-to-end latency. Session- batched writing collapses A-Mem's per-turn writer loop (4931 calls) into per-session extraction (450 calls), and the subject/predicate gate restricts linker invocations to 1107 total – only candidate pairs whose subjects or predicates lexically overlap. The result is -63.5% of A-Mem's token budget at comparable latency. Against Flat RAG, which has no writer, linker, or parser, TTMG costs $\sim 12\times$ more tokens per question; this is the intrinsic cost of maintaining a claim-level memory with truth-maintenance edges, bought in exchange for the abstention, supersede, and temporal semantics quantified in §4.2 and §5. Deployment cost can be further reduced by reusing the writer's output across multiple downstream questions sharing the same conversation haystack.

5 Analysis and Ablations

To isolate which TTMG mechanism drives the temporal-reasoning gain, we run the **–contradict** ablation: we keep the claim schema, the session-batched writer, the temporal-anchor parser, and the maximum-consistent-subgraph retriever, but disable the LLM linker so no `contradict` or `supersede` edges are ever written. Ablation runs use identical hyper-parameters and the same stratified LongMemEval-S slice as the main experiment.

Paired dominance on abstention and SSU. A paired per-question analysis on the 9 abstention questions shows that TTMG *strictly dominates* Flat RAG: 4 questions are answered correctly by TTMG and incorrectly by Flat RAG, 0 in the other direction (both correct: 4; neither: 1). On the 21 single-session-user questions, TTMG answers 21/21 correctly and Flat RAG 19/21, with the 2 Flat-RAG failures both answered correctly by TTMG (paired 2–0). Every strict-dominance win

Table 3: Ablation: removing only the LLM linker (the single source of `contradict`/`supersede` edges) drops *every* measured slice, confirming its causal role. Numbers are accuracy (%) on a 150-question stratified LongMemEval-S slice.

Method	**Overall** n=150	**KU** n=23	**TR** n=40	**ABS** n=9	**SSU** n=21
Flat RAG (strong baseline)	68.0	78.3	52.5	44.4	90.5
TTMG (full)	61.3	78.3	52.5	88.9	100.0
TTMG **–contradict**	58.7	69.6	47.5	66.7	95.2
drop vs full	-2.6	-8.7	-5.0	-22.2	-4.8

TTMG reports comes from a specific baseline failure on the same question, not from different sampled subsets.

5.1 What the ablation tells us

The **–contradict** variant keeps the claim schema, the session-batched writer, the retriever, and the hybrid reader context intact; it disables only the LLM linker. Every measured slice drops. The biggest drop is on **abstention** (-22 pp, 88.9% $\rightarrow$ 66.7%): without `contradict` edges, the reader cannot tell that two candidate claims disagree and instead hallucinates an answer. **Knowledge-update** loses 9 pp (78 % $\rightarrow$ 70 %): without `supersede` edges, retrieval returns the stale claim alongside the fresh one. **Temporal reasoning** loses 5 pp. **Single-session-user** loses 5 pp. **Overall** loses 2.6 pp. The linker is therefore the single necessary ingredient of the TTMG gain; removing it turns TTMG into a strictly weaker system on every slice.

The write-time overhead of the linker is small: a cheap subject/predicate lexical check gates the LLM call, so it fires on a minority of new claims, and its incremental token cost is strictly below A-Mem's per-turn note-analysis cost (§4.3).

5.2 Regression taxonomy (SSA, SSP, MS)

Standalone TTMG regresses on three slices (SSA, SSP, MS). Inspecting the paired errors (questions Flat answers correctly and TTMG does not) reveals a single dominant mechanism: *over-abstention* driven by the claim-level retrieval surface. We categorise the 18 Flat-only-correct paired errors on these slices into three failure modes.

F1. Writer entity gap (SSA, 7/7). Assistant-authored questions such as "remind me of the two companies you mentioned that prioritise employee safety" require the retriever to surface the assistant's own prior answer. The session-batched writer extracts user-side claims more densely than assistant-side ones, so the needed entity (*Patagonia*, *Veja*, *2014* construction year) is not in the claim graph. The retriever returns only loose neighbours; the reader, asked to answer "only from claims," refuses. Flat RAG surfaces the raw assistant turn verbatim and answers correctly.

F2. Preference reasoning (SSP, 3/3). Preference questions ("any cocktail suggestions?", "any documentary recommendations?") require the reader to combine a stated preference with a mild recommendation; the claim schema's (s, p, o) triples carry the fact but not the reasoning pattern. TTMG's reader abstains because no single claim answers the question; Flat RAG's raw-turn context lets the reader reason implicitly.

F3. Aggregation across sessions (MS, 8/8). Multi-session questions such as "how much more expensive was the taxi ride than the train fare?" require retrieving *two* related claims and subtracting. TTMG's top-k_{keep} retrieval returns one; the consistency filter may drop the other if their validity intervals overlap differently; the reader abstains. Flat RAG, with raw-turn context spanning both sessions, computes the difference.

These failure modes are all *structural* to the claim-as-atom abstraction and motivate the router composition (§4.2): on F1–F3-shaped questions, fall back to Flat RAG. They also suggest two direct method extensions – denser assistant-side extraction (F1) and multi-claim composition in the reader prompt (F3) – that we leave as future work rather than addressing post-hoc in this submission.

5.3 Qualitative success example

On the "last Friday" artist question, a flat hybrid RAG hallucinates a band name when the user's own message only described the band by instrumentation; TTMG, seeing a claim with matching `valid_from` and the `active`/`past` flags, refuses to invent one. Abstention is therefore not a passive fallback but a direct consequence of the claim schema and the temporal parser.

6 Conclusion

We addressed the problem of maintaining factual truth in long-context conversational memory. The task is not merely to retrieve textually similar past utterances, but to ensure that the information presented to the model over time remains consistent with the evolving ground truth, even as user statements are corrected or superseded.

Our solution introduces a structured memory mechanism that explicitly models the truth status of information. Each memory unit is a claim schema annotated with its validity and linked to others via *contradict* and *supersede* edges. This graph enables truth-consistent retrieval, where only currently valid claims are surfaced, and strategic abstention when the memory lacks sufficient information to answer reliably. This approach directly enforces a coherence constraint over the memory's contents.

Empirically, TTMG matches Flat RAG on outcome accuracy at statistical parity on both overall LongMemEval-S (p=0.21) and the full abstention set (p=1.00, n=30), while delivering -63.5% of A-Mem's token cost and providing an explicit abstention policy that fires on 80.0% of abstention questions (versus 0% for Flat). On single-session-user questions TTMG captures every Flat correct answer and two more (paired 2–0, n=21). Composing TTMG with Flat RAG at inference time via our TTMG-Abstain (67.3%) or TTMG+R-LLM (66.0%) routers recovers overall accuracy while preserving TTMG's supersede behaviour wherever the router fires. The oracle upper bound of 82.0% documents that claim-level and raw-text retrieval carry substantially complementary information; harvesting that complementarity beyond the feature router is a direct target for future work. Explicitly reasoning about truth maintenance is therefore a useful primitive to add to long-running conversational agents, particularly where abstention and user-fact fidelity matter more than bulk retrieval accuracy.

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A Prompt templates

All TTMG LLM calls route through the same chat interface (OpenAI-compatible). The three module-specific prompts below are what the writer, linker, and retriever query-parser actually see; small differences between the session-batched writer and the single-turn writer are flagged.

A.1 Session-batched writer

> *System:* You extract factual claims from conversation turns for a long-term memory system. Only extract information the speaker ACTUALLY asserts. Do not invent content. Respond with strict JSON only.
>
> *User:* Extract factual claims from the conversation below. Session metadata: `session_id`, `session_ts`. Turns are numbered. For each turn containing factual content (facts, preferences, plans, states), output claims with fields `turn_id`, `speaker`, `content`, `subject`, `predicate`, `object`, `valid_from`, `valid_to`, `polarity` $\in \{\texttt{assert}, \texttt{deny}\}$, `volatility` $\in \{\texttt{stable}, \texttt{preference}, \texttt{state}\}$, `confidence` $\in [0, 1]$. Skip greetings, small talk and pure questions asked of the model. Keep content under 25 words. Return `{"claims":[...]}`.

A.2 Link labeller

> *System:* You label pairs of factual claims as one of `support`, `contradict`, `supersede`, or `unrelated`. Respond in strict JSON only.
>
> *User:* For the NEW claim and each CANDIDATE claim output a label and a confidence in $[0, 1]$. **support**: same fact; both coexist. **contradict**: disagree on the same (s, p) at overlapping times. **supersede**: the NEW claim replaces the CANDIDATE (same (s, p), later `valid_from`, or explicit correction). **unrelated**: otherwise. Be conservative: default to `unrelated` unless subject and predicate match. Return `{"pairs":[{"id":...,"label":...,"confidence":...}]}`.

A.3 Query parser

> *System:* You parse a question asked of a long-term memory system. Respond with strict JSON only.
>
> *User:* Parse the question. Return fields `temporal_anchor` (time string or `null`), `asks_history` (question asks about a past state that may have been superseded), `asks_current` (question asks about the current state), `entity` (main subject or `null`), `intent` $\in \{\texttt{lookup}, \texttt{update\_check}, \texttt{temporal\_reasoning}, \texttt{small\_talk}\}$.

Table 4: TTMG default hyper-parameters, used unchanged across all benchmarks.

Component	Value / model
Embedding model	`all-MiniLM-L6-v2`
Reader (answer generation)	`deepseek-v3.2`
Writer / linker / parser	`Kimi-K2`
Intent classifier (TTMG+R-LLM)	`Kimi-K2`, zero-shot
Retrieval k	8
k_{keep} at read time	3
Linker candidate k at write time	3
Similarity gate for linker	0.65
Bypass threshold	0.85
Hard-edge threshold τ_{hard}	0.7
Minimum claim-query similarity	0.35
Session-batched writer	on
Max sessions ingested per question	3
Evaluation seed	0 (stratified)

B Hyper-parameters and inference-time router

B.1 LLM intent classifier for TTMG+R-LLM

The TTMG+R-LLM router uses a one-shot Kimi-K2 call per incoming question to predict one of the six LongMemEval types. The classifier prompt lists the six labels with one example each and asks for a single-key JSON response. On the 150-question stratified evaluation slice, its overall accuracy is 64.7%. Per-class recall is: `single-session-user` 81.0%, `single-session-assistant` 100.0%, `temporal-reasoning` 87.5%, `multi-session` 42.5%, `knowledge-update` 39.1%, `single-session-preference` 22.2%. The high recall on SSU and SSA is what makes the simple "route-SSU-to-TTMG-else-Flat" rule practical; the lower recall on MS and KU is the dominant source of the gap to the oracle upper bound and a natural target for a better-trained classifier.

B.2 Reproducibility statement

We release the full source (writer/linker/parser/reader prompts, the retriever algorithm, the claim schema, and all experiment scripts), the N=150 stratified subsample identifiers and seed (`seed=0`, stratified on `question_type`), all hyper-parameters (Table 4), and the raw per-question outputs (`results/pilot_n150_*.json`), which includes the `question_id`, `correct` flag, and metric counters per question. McNemar p-values and bootstrap CIs are computed by `scripts/analyze_n150.py`. The LLM intent classifier is implemented in `scripts/llm_intent_classifier.py`. All calls route through the MAAS OpenAI-compatible endpoint; model identifiers (`deepseek-v3.2`, `Kimi-K2`) are frozen strings provided by the hosting platform.

C Dataset details

LongMemEval-S contains 500 questions across six types: *single-session-user* (70), *single-session-assistant* (56), *single-session-preference* (30), *multi-session* (133), *knowledge-update* (78), and *temporal-reasoning* (133). A further 30 questions bearing an `_abs` suffix in their question identifier are *abstention* probes: the reference answer states that the information was never mentioned, so the correct response is "I do not know". Each question is paired with an average of $\sim$53 haystack sessions drawn from a user's conversation history; `answer_session_ids` labels the session(s) from which the answer may be derived.

Each of the 50 questions retrieves memory from its own haystack; we cap the ingested haystack at the 3 highest-priority sessions (the answer session plus the two earliest matching sessions) to keep the per-question ingestion budget manageable while preserving the ratio of relevant-to-irrelevant sessions the LongMemEval benchmark intends.